

CYBERSECURITY & ETHICAL HACKING INTERNSHIP

TASK 1 — FOUNDATION & ENVIRONMENT SETUP

Lab Setup Report

Submitted to:

ApexPlanet Software Pvt. Ltd.

Internship Domain:

Cybersecurity & Ethical Hacking

Task:

Task 1 — Foundation & Environment Setup

Tools & Technologies:

VirtualBox | Kali Linux | Metasploitable2 | Wireshark | OpenSSL | Nmap | Burp Suite | Netcat

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Section 2 — Introduction

1. Introduction

The objective of Task 1 is to build strong fundamentals in cybersecurity, networking, cryptography, and establish a professional cybersecurity laboratory environment.

As part of this task, a virtual cybersecurity lab was configured using VirtualBox. Kali Linux was used as the attacker/security-testing machine, while Metasploitable2 was configured as the target machine. A Host-Only network was used to allow communication between the virtual machines within an isolated laboratory environment.

The task also involved learning fundamental Linux commands, networking concepts, cryptography concepts, and familiarization with commonly used cybersecurity tools such as Wireshark, Nmap, Burp Suite, and Netcat. The internship document specifically identifies these areas as part of Task 1.

2. Objectives

The main objectives of this task were:

- To understand fundamental cybersecurity concepts.
- To understand the CIA Triad and common cyber threats.
- To set up an isolated cybersecurity laboratory.
- To configure Kali Linux and Metasploitable2 in VirtualBox.
- To establish communication between the virtual machines using a Host-Only network.
- To understand basic Linux commands and permissions.

- To study fundamental networking concepts.
- To perform basic encryption and decryption using OpenSSL.
- To become familiar with Wireshark, Nmap, Burp Suite, and Netcat.
- To document the laboratory setup and practical activities.

Cybersecurity Fundamentals

3.1 CIA Triad

The CIA Triad is a fundamental cybersecurity model consisting of **Confidentiality, Integrity, and Availability**. It represents three important objectives for protecting information and systems.

Confidentiality:

Confidentiality ensures that information is accessible only to authorized users. Security controls such as authentication, access control, and encryption help protect sensitive information from unauthorized access.

Integrity:

Integrity ensures that information remains accurate, complete, and protected from unauthorized modification. Hashing, digital signatures, and access controls can be used to help maintain data integrity.

Availability:

Availability ensures that systems and information are accessible to authorized users when required. Measures such as backups, redundancy, and protection against denial-of-service attacks can help maintain availability.

3.2 Common Threat Types

The Task 1 syllabus identifies the following common cybersecurity threats:

Phishing:

A social-engineering technique in which an attacker attempts to deceive users into revealing information or performing an unsafe action, commonly through fraudulent messages or websites.

Malware:

Malicious software designed to perform unauthorized or harmful activities on a computer system.

DDoS:

Distributed Denial-of-Service attacks attempt to overwhelm a service or system with traffic from multiple sources, affecting its availability.

SQL Injection:

An attack technique involving malicious SQL input intended to manipulate a vulnerable application's database queries.

Brute Force:

An attack in which repeated authentication attempts are made using different passwords or credentials.

Ransomware:

Malicious software that can restrict access to data or systems and is commonly associated with demands for payment from victims.

3.3 Attack Vectors

The Task 1 material identifies three attack vectors: **Social Engineering, Wireless Attacks, and Insider Threats.**

Social Engineering:

Uses deception or psychological manipulation to persuade people to reveal information or perform actions that may compromise security.

Wireless Attacks:

Target wireless networks or communications to gain unauthorized access, intercept traffic, or disrupt connectivity.

Insider Threats:

Security risks originating from individuals within an organization who have legitimate access to systems or information.

3.4 Importance of Cybersecurity

Cybersecurity helps protect information systems, networks, applications, and data from unauthorized access, modification, disruption, and other threats. Understanding fundamental security principles provides a foundation for identifying vulnerabilities and applying appropriate security controls.

Lab Environment & Virtual Machine Setup

4.1 Virtualization Environment

VirtualBox was used as the virtualization platform for creating and managing the cybersecurity laboratory. The lab consists of an attacker/security-testing machine and a deliberately vulnerable target machine.

The laboratory environment was configured as follows:

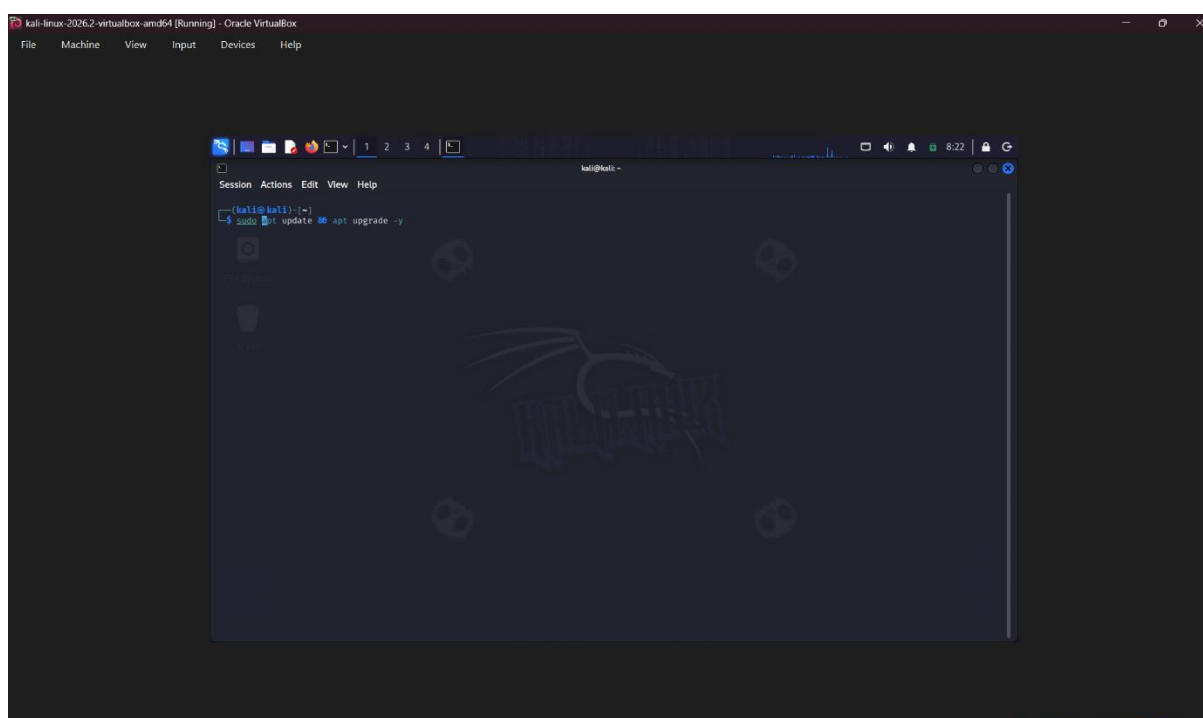
Component	Configuration
Virtualization Platform	Oracle VirtualBox
Attacker Machine	Kali Linux
Target Machine	Metasploitable2
Network Type	Host-Only Adapter
Kali IP Address	192.168.56.101
Metasploitable2 IP Address	192.168.56.102

The Host-Only network allows the virtual machines to communicate with each other within the isolated laboratory environment.

4.2 Kali Linux

Kali Linux was configured as the primary security-testing machine. It was used during the practical exercises to perform network testing, packet capture, cryptographic operations, and security-tool familiarization.

Figure 1: Kali Linux Running



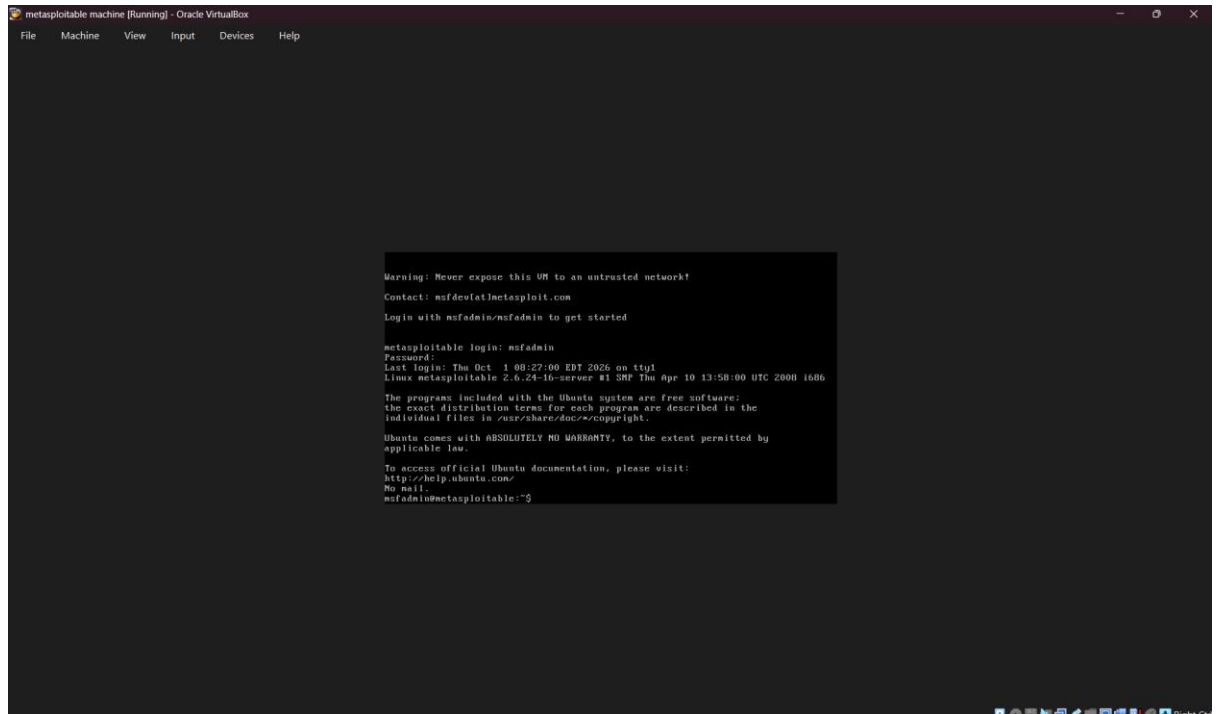
Caption:

Figure 1 — Kali Linux security-testing environment running in VirtualBox.

4.3 Metasploitable2

Metasploitable2 was configured as the target machine for the cybersecurity laboratory. It provides a controlled and intentionally vulnerable environment for security testing and learning.

Figure 2: Metasploitable2 Running



Caption:

Figure 2 — Metasploitable2 target machine running in VirtualBox.

4.4 Host-Only Network Configuration

A Host-Only Adapter was configured to provide communication between Kali Linux and Metasploitable2 while keeping the practical activities within the laboratory network.

The configured addresses were:

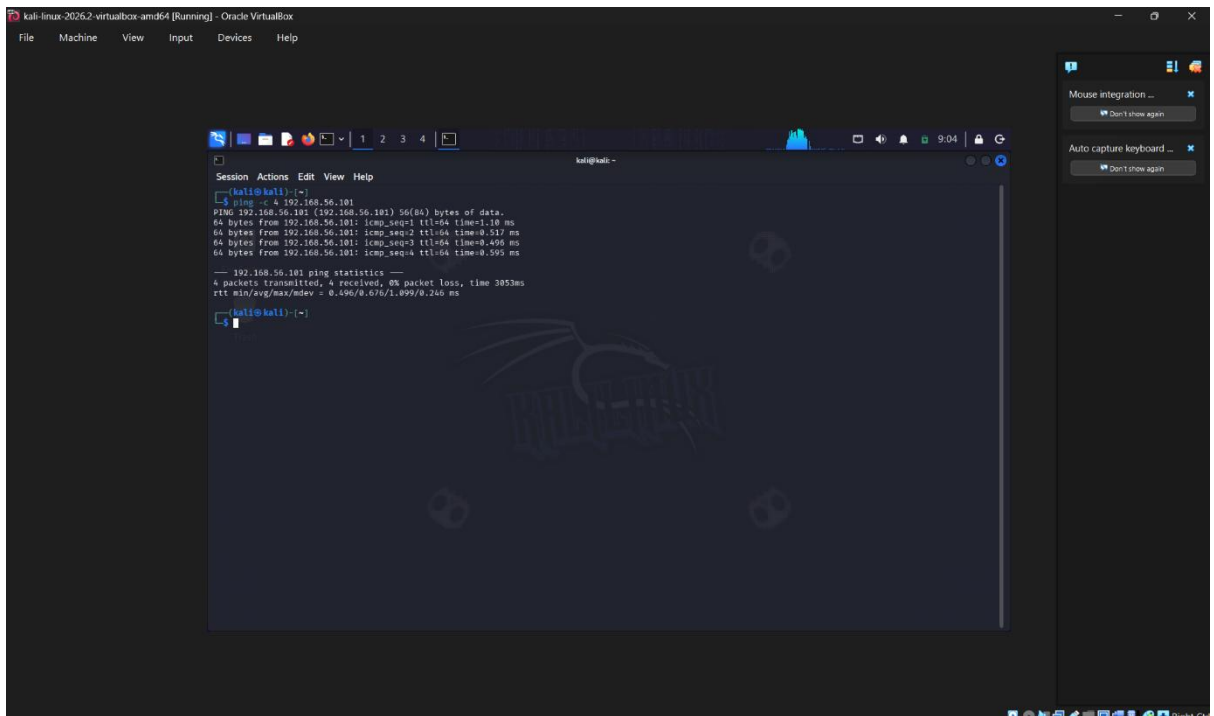
- **Kali Linux:** 192.168.56.101
- **Metasploitable2:** 192.168.56.102

Connectivity was verified from Kali Linux using:

```
ping -c 4 192.168.56.102
```

The successful responses confirmed communication between the two virtual machines.

Figure 3: Host-Only Network Configuration



Caption:

Figure 3 — Host-Only network configuration for the cybersecurity laboratory.

4.5 Network Connectivity Verification

The connection between Kali Linux and Metasploitable2 was tested using ICMP ping. Successful echo replies were received from the Metasploitable2 IP address, confirming that both machines could communicate over the configured Host-Only network.

Linux Fundamentals

The Task 1 syllabus covers **file-system navigation, permissions, package management, and networking commands** as part of Linux fundamentals.

5.1 File-System Navigation

Linux provides command-line utilities for navigating and working with the file system.

pwd — Print Working Directory

Displays the current working directory.

pwd

ls — List Directory Contents

Displays files and directories in the current location.

ls

cd — Change Directory

Used to move between directories.

cd /tmp

The commands were practiced in Kali Linux to understand basic file-system navigation.

5.2 File Permissions

Linux uses permissions to control access to files and directories.

The chmod command is used to modify file permissions.

Example:

chmod 600 testfile.txt

The chown command is used to change the owner of a file or directory.

chown user:group testfile.txt

File permissions can be viewed using:

ls -l

5.3 Package Management

Kali Linux uses Debian-based package management.

The main package-management tools covered in the task are:

APT:

apt

APT is used to install, update, and manage software packages.

DPKG:

dpkg

DPKG is a lower-level package-management utility for Debian packages.

5.4 Networking Commands

Several command-line utilities are useful for examining network configuration and connectivity.

ifconfig

Displays network-interface information.

ifconfig

ping

Tests network connectivity between systems.

`ping -c 4 192.168.56.102`

netstat

Displays network connections and related information.

netstat

traceroute

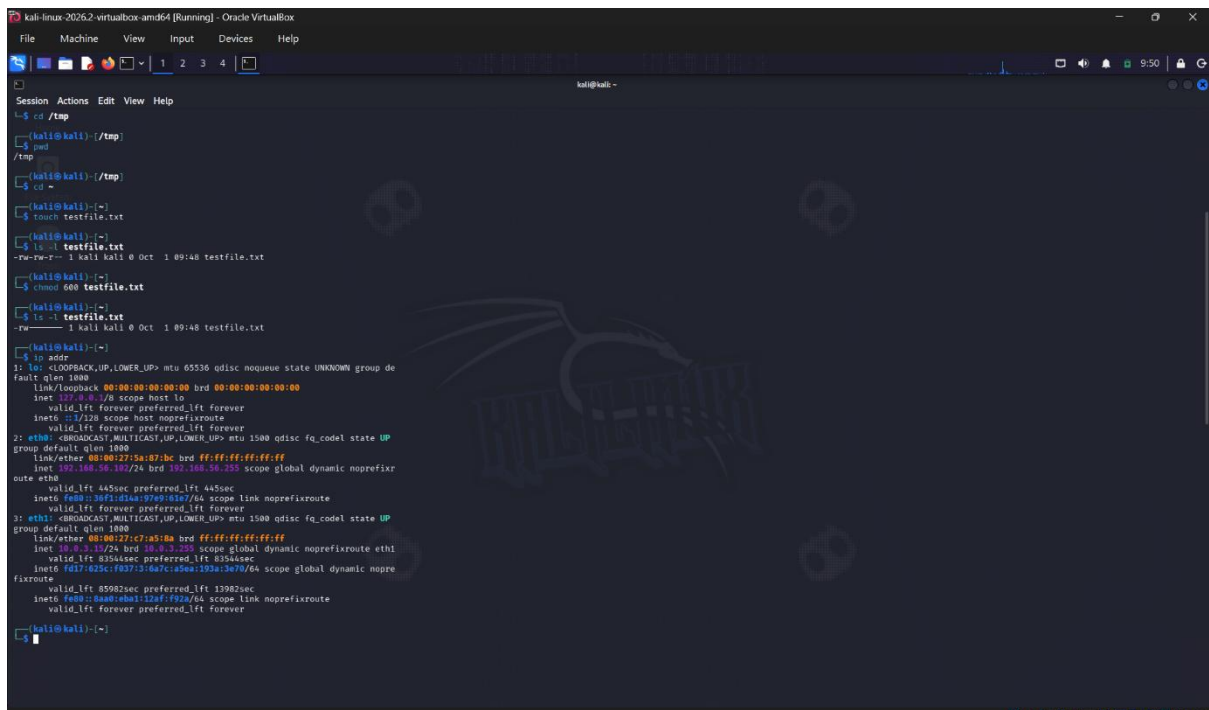
Shows the network path taken toward a destination.

`traceroute <destination>`

These commands provide basic familiarity with Linux networking and system administration.

5.5 Practical Evidence

The Linux commands were executed in Kali Linux during the laboratory setup.



```
kali-linux-20262-virtualbox-amd64 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

kali@kali:~$ cd /tmp
kali@kali:~/tmp$ touch testfile.txt
kali@kali:~/tmp$ ls -l testfile.txt
-rw-r--r-- 1 kali kali 0 Oct 1 09:48 testfile.txt
kali@kali:~/tmp$ du -sh testfile.txt
4.0K testfile.txt
kali@kali:~/tmp$ ls -l testfile.txt
-rw-r--r-- 1 kali kali 0 Oct 1 09:48 testfile.txt
kali@kali:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group de
fault qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP
    group default qlen 1000
    link/ether 08:00:27:5a:b7:bc brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.102/24 brd 192.168.56.255 scope global dynamic noprefixr
oute eth0
        valid_lft 445sec preferred_lft 445sec
    inet6 fe80::3a5f:d2a8:5f5d:1377/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP
    group default qlen 1000
    link/ether 08:00:27:c7:a5:8a brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.12/24 brd 192.168.1.255 scope global dynamic noprefixroute eth1
        valid_lft 8354sec preferred_lft 8354sec
    inet6 fd17:623c:f037:3:6a7c:a5e4:193a:3e70/64 scope global dynamic nopr
efixroute
        valid_lft 85982sec preferred_lft 13982sec
    inet6 fe80::d9a0:eb01:12af:f92/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
kali@kali:~$
```

Caption:

Figure 4 — Linux navigation, permissions, and networking commands demonstrated in Kali Linux.

Networking Fundamentals

The Task 1 material covers the **OSI Model, TCP/IP Protocol Suite, DNS, HTTP/HTTPS, IP addressing, subnetting, and NAT** as networking fundamentals.

6.1 OSI Model

The **Open Systems Interconnection (OSI) Model** divides network communication into seven conceptual layers:

Layer Name	Main Function
7	Application Provides network services to applications
6	Presentation Handles data representation and formatting
5	Session Establishes and manages communication sessions
4	Transport Provides end-to-end data delivery
3	Network Handles logical addressing and routing
2	Data Link Handles frames and local network communication
1	Physical Transmits data as physical signals

Understanding these layers helps in analyzing network communication and troubleshooting network-related security issues.

6.2 TCP/IP Protocol Suite

The TCP/IP model describes the protocols used for communication across modern computer networks.

Important protocols include:

- **TCP** — Provides reliable, connection-oriented communication.
- **UDP** — Provides connectionless communication with lower overhead.
- **IP** — Provides logical addressing and routing.
- **HTTP/HTTPS** — Used for web communication.
- **DNS** — Used for resolving domain names to IP addresses.

6.3 DNS

Domain Name System (DNS) translates human-readable domain names into IP addresses.

For example:

example.com → IP address

This allows users to access network services using domain names instead of remembering numerical IP addresses.

6.4 HTTP and HTTPS

HTTP (Hypertext Transfer Protocol) is used for communication between web clients and servers.

HTTPS (HTTP Secure) uses encryption through TLS to protect communication between the client and server.

HTTPS helps provide:

- Confidentiality
 - Data integrity
 - Server authentication through digital certificates
-

6.5 IP Addressing

An IP address identifies a device or network interface on an IP network.

In the laboratory environment:

Kali Linux → 192.168.56.101

Metasploitable2 → 192.168.56.102

These addresses allowed the two virtual machines to communicate over the Host-Only network.

6.6 Subnetting

Subnetting divides an IP network into smaller logical networks.

It can be used to:

- Organize network devices.
- Manage IP address allocation.
- Separate network segments.
- Improve network management.

The Host-Only laboratory network provides a controlled network segment for communication between the virtual machines.

6.7 NAT

Network Address Translation (NAT) translates IP addresses between different networks.

NAT is commonly used when devices on a private network access external networks through a shared public IP address.

In this laboratory, the **Host-Only Adapter** was used for communication between the Kali and Metasploitable2 machines.

6.8 Networking Practical

Connectivity between the two virtual machines was verified using ICMP:

```
ping -c 4 192.168.56.102
```

The successful replies confirmed network communication between Kali Linux and Metasploitable2.

Cryptography & OpenSSL

The Task 1 syllabus covers **symmetric and asymmetric encryption, hashing, MD5, SHA-256, digital certificates, SSL/TLS**, and a practical exercise to **encrypt and decrypt messages using OpenSSL**.

7.1 Symmetric Encryption

Symmetric encryption uses the **same key** for encryption and decryption.

It is generally suitable for efficiently protecting data when the communicating parties can securely share the encryption key.

Example:

Plaintext + Secret Key → Encryption → Ciphertext

Ciphertext + Secret Key → Decryption → Plaintext

7.2 Asymmetric Encryption

Asymmetric encryption uses a **key pair**:

- Public key
- Private key

The public key can be shared, while the private key is kept secret.

Asymmetric cryptography is commonly associated with secure key exchange, authentication, and digital signatures.

7.3 Hashing

Hashing converts input data into a fixed-length value called a **hash**.

Unlike encryption, hashing is designed as a one-way operation rather than something that can simply be decrypted to recover the original data.

Hashing can be used to help verify data integrity.

7.4 MD5

MD5 (Message-Digest Algorithm 5) is a hashing algorithm that produces a 128-bit hash value.

MD5 is included in the Task 1 cryptography topics for understanding hashing.

7.5 SHA-256

SHA-256 is a member of the SHA-2 family of cryptographic hash functions and produces a 256-bit hash value.

It can be used to generate a digest that helps verify whether data has been modified.

7.6 Digital Certificates

Digital certificates are used to associate a public key with an identity. They are an important component of public-key infrastructure and are used in secure communications.

7.7 SSL/TLS

TLS (Transport Layer Security) is used to protect network communication by providing encryption and helping provide authentication and integrity.

HTTPS uses TLS to secure web communication.

7.8 OpenSSL Practical

As part of the practical exercise, a text message was created and encrypted using OpenSSL with **AES-256-CBC**.

The message was first stored in a text file and then encrypted:

```
openssl enc -aes-256-cbc -salt -pbkdf2 -in message.txt -out message.enc
```

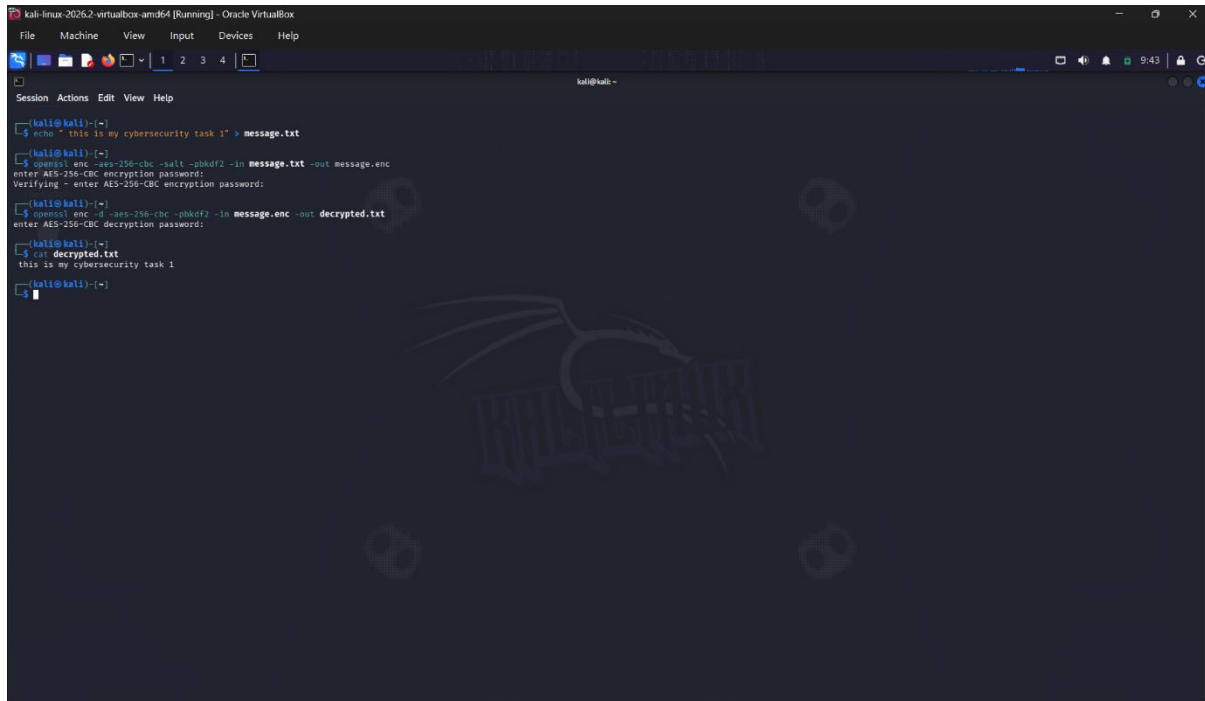
The encrypted file was subsequently decrypted using:

```
openssl enc -d -aes-256-cbc -pbkdf2 -in message.enc -out decrypted.txt
```

The decrypted message was verified using:

```
cat decrypted.txt
```

The original message was successfully recovered after decryption.



```
kali-linux-20262-virtualbox-amd64 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
1 2 3 4
kali@kali:~$ echo "this is my cybersecurity task 1" > message.txt
kali@kali:~$ openssl enc -aes-256-cbc -salt -pbkdf2 -in message.txt -out message.enc
enter AES-256-CBC encryption password:
Verifying - enter AES-256-CBC encryption password:
kali@kali:~$ openssl enc -d -aes-256-cbc -pbkdf2 -in message.enc -out decrypted.txt
enter AES-256-CBC decryption password:
kali@kali:~$ cat decrypted.txt
this is my cybersecurity task 1
kali@kali:~$
```

Caption:

Figure 5 — OpenSSL encryption and decryption practical performed in Kali Linux.

Security Tool Familiarization

Task 1 identifies **Wireshark, Nmap, Burp Suite, and Netcat** for tool familiarization.

8.1 Wireshark

Wireshark is a network packet-analysis tool used to capture and inspect network traffic.

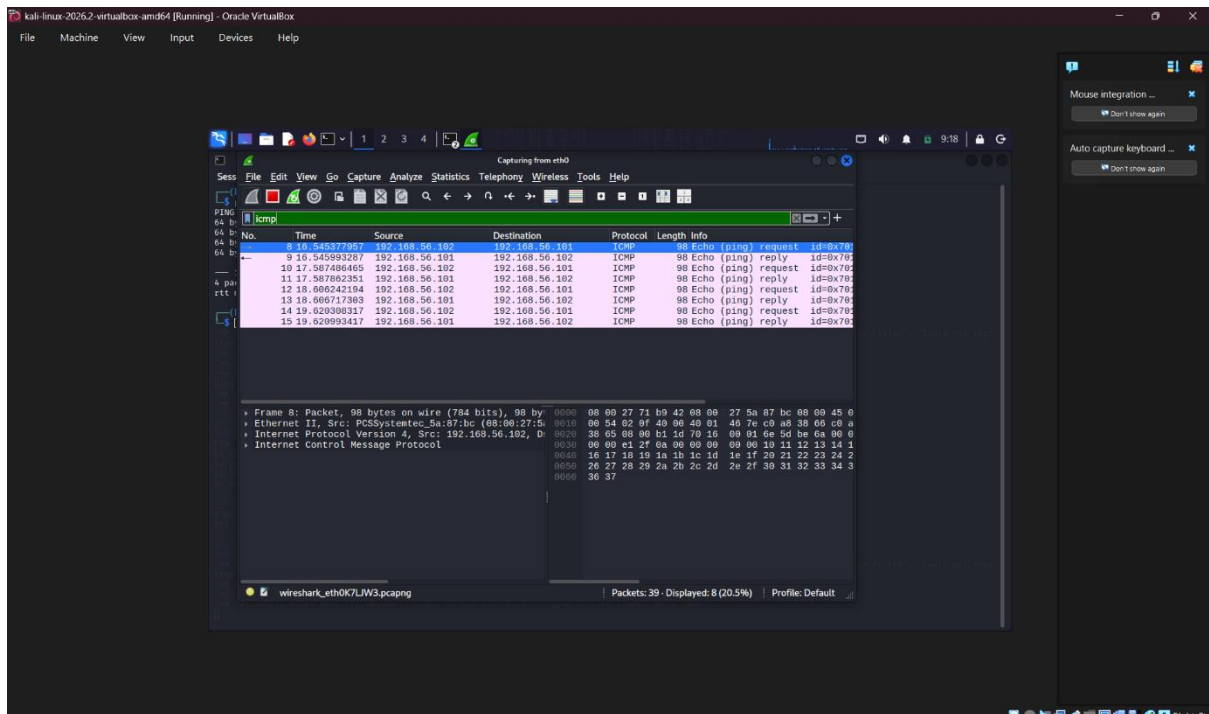
In this laboratory, Wireshark was used to capture the ICMP traffic generated while testing connectivity between Kali Linux and Metasploitable2.

The capture showed communication between:

Kali Linux → 192.168.56.101

Metasploitable2 → 192.168.56.102

The icmp display filter was used to isolate the relevant packets.



Caption:

Figure 6 — Wireshark capture showing ICMP communication between Kali Linux and Metasploitable2.

8.2 Nmap

Nmap is a network-scanning tool used to discover hosts and identify services available on a network.

A basic scan was performed against the Metasploitable2 laboratory machine:

```
nmap 192.168.56.102
```

A service-version scan was then performed:

```
nmap -sV 192.168.56.102
```

The scan provided information about open ports and services running on the target machine.

```

kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

kali@kali:~$ nmap -sV 192.168.56.101
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-01 09:45 -0400
Nmap scan report for 192.168.56.101
Host is up (0.0000s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache/2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind      2 (RPC #100000)
139/tcp   open  smb          Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  smb          Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rshcd
513/tcp   open  login        OpenSSH or Solaris rlogind
514/tcp   open  shell        Netkit rshd
1099/tcp  open  java-rmi     GNU Classpath gmicregistry
1524/tcp  open  bindshell    Metasploitable root shell
2049/tcp  open  nfs          2.4 (RPC #100003)
2222/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
5208/tcp  open  vnc           VNC (protocol 3.3)
6000/tcp  open  x11          (access denied)
6607/tcp  open  irc          UnrealIRCd
8080/tcp  open  http         Apache/2.2.8 (Ubuntu)
8180/tcp  open  http         Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 08:00:27:1B:89:42 (Oracle VM VirtualBox virtual NIC)
Service Info: Hosts: metasploitable.localdomain, irc, metasploitable.LAN; OS: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 12.15 seconds

kali@kali:~$

```

Caption:
Figure 7 — Nmap service and port scan of the Metasploitable2 laboratory target.

8.3 Burp Suite

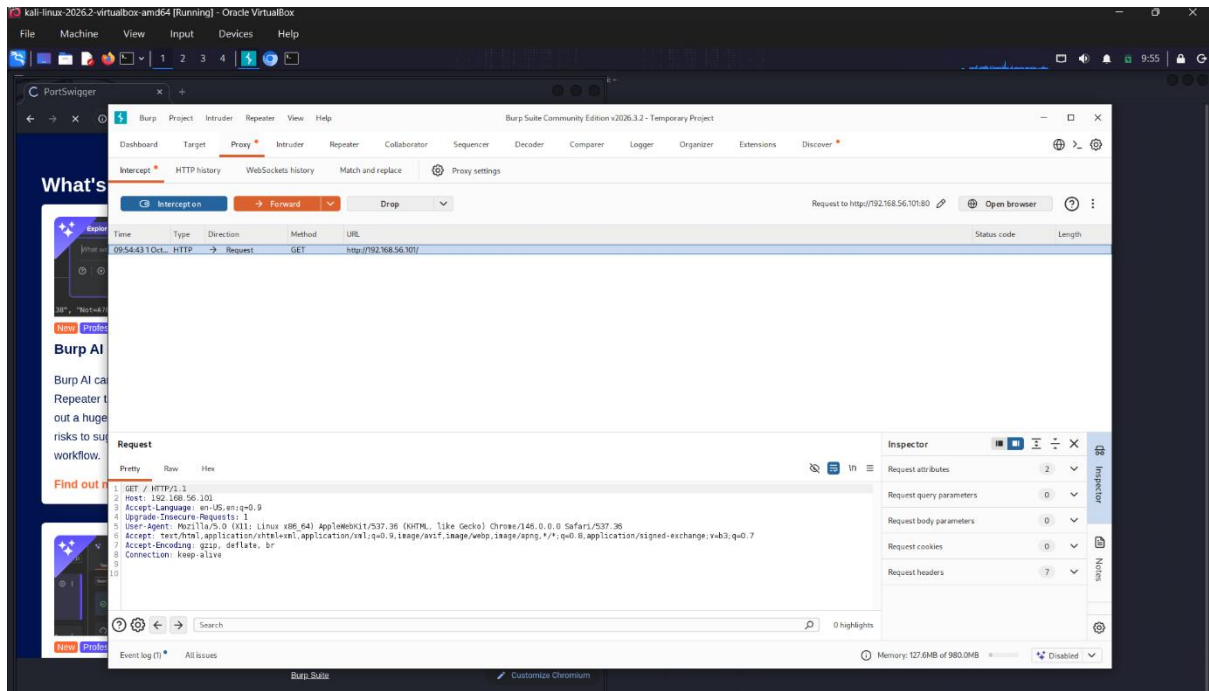
Burp Suite is a web-security testing platform that can act as an intermediary between a browser and a web application.

During the practical exercise, Burp Suite's **Proxy/Intercept** functionality was used to observe an HTTP request sent to the Metasploitable2 web server.

An intercepted request included information such as:

GET / HTTP/1.1

Host: 192.168.56.102



Caption:
Figure 8 — Burp Suite intercepting an HTTP request from the laboratory web application.

8.4 Netcat

Netcat is a networking utility that can be used for network communication and debugging.

A basic local communication test was performed using port 4444.

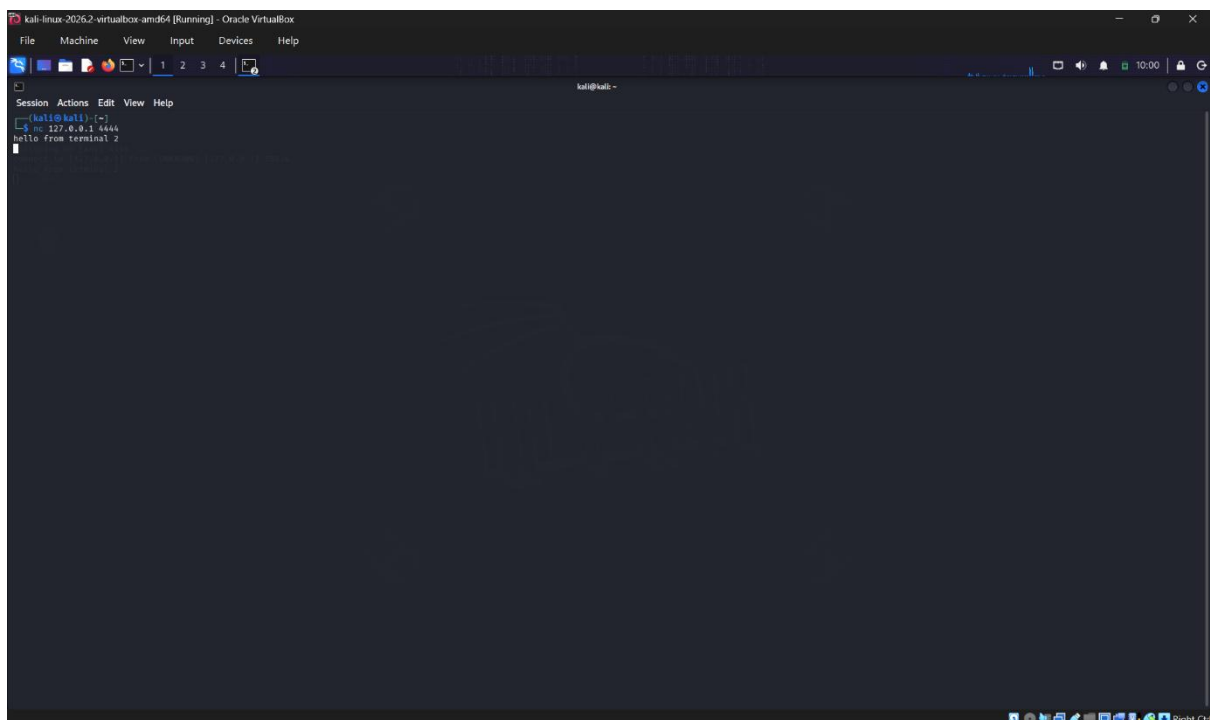
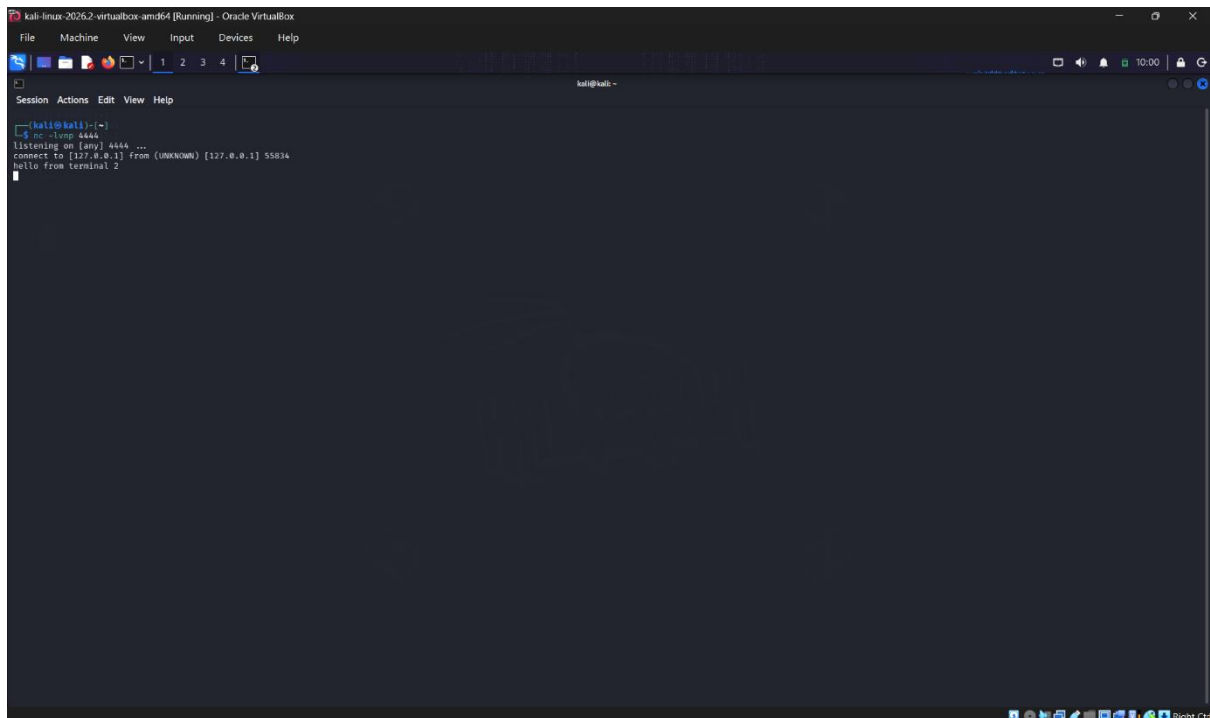
The listener was started with:

```
nc -lvp 4444
```

A second terminal connected to the listener:

```
nc 127.0.0.1 4444
```

A test message was sent between the two terminals to verify the connection.



Caption:

Figure 9 — Netcat local communication test using port 4444.

8.5 Summary of Tools

Tool	Purpose in the Lab
Wireshark	Packet capture and network traffic analysis

Tool	Purpose in the Lab
Nmap	Network and service scanning
Burp Suite	Web request interception
Netcat	Network communication and debugging

Results, Evidence & Conclusion

9.1 Lab Setup Results

The cybersecurity laboratory environment was successfully established using VirtualBox. Kali Linux was configured as the security-testing machine and Metasploitable2 as the target machine. Both systems were connected through a Host-Only network.

Connectivity between the two machines was verified using ICMP ping, confirming communication within the laboratory network.

9.2 Practical Activities Completed

The following practical activities were performed during Task 1:

Activity	Status
VirtualBox laboratory setup	Completed
Kali Linux setup	Completed
Metasploitable2 setup	Completed
Host-Only network configuration	Completed
Kali ↔ Metasploitable connectivity test	Completed
Wireshark packet capture	Completed
OpenSSL encryption/decryption	Completed
Nmap scanning	Completed
Linux fundamentals	Completed
Burp Suite familiarization	Completed
Netcat communication test	Completed

9.3 Evidence Collected

The following screenshots were collected as practical evidence:

1. Kali Linux running
2. Metasploitable2 running
3. Host-Only network configuration
4. Wireshark ICMP capture
5. OpenSSL encryption/decryption
6. Nmap scan
7. Linux fundamentals
8. Burp Suite interception
9. Netcat communication

These screenshots document the practical activities performed during the laboratory setup.

9.4 Conclusion

Task 1 provided a foundation in cybersecurity concepts, networking, cryptography, Linux fundamentals, and security-tool usage.

A controlled cybersecurity laboratory was established using VirtualBox, Kali Linux, and Metasploitable2. Network communication was verified through the Host-Only configuration, and practical exercises were performed using Wireshark, OpenSSL, Nmap, Burp Suite, and Netcat.

The completed laboratory provides an isolated environment for carrying out subsequent cybersecurity learning and testing activities.